SECTION 000500 – FENN BUNKHOUSE AND COOKHOUSE ROOFING

1.1 GENERAL

This item shall consist of installing a new roof on the Fenn Bunkhouse and Cookhouse on the Nez Perce-Clearwater National Forests. Work includes but is not limited to removing, replacing roofing, repairing sub roof assemblies, installing weather proofing, ice and water shield, cedar breather, cedar saver, and water barriers. As a significant historical resource(s) it is imperative that maintenance activities DO NOT ADVERSELY AFFECT the historical integrity of the two structures. Accordingly, all materials or replacement materials used for maintenance must be "in kind" (e.g. use of original wall and trim paint colors, replacement with matching style trim and wall boards, etc.). If there are any questions about acceptable materials or techniques to be used, please contact the Contracting Officer. Fenn attic spaces must remain sealed throughout entire construction project to prevent entry from bats and other wildlife.

Any and all excess materials and property paid for will be retained by the Forest Service. Arranging a storage location for excess material will be at the discretion of the Contracting Officer.

The project is located at Fenn Ranger Station, this site is accessible year around.

Sub roof assemblies may not require repair and are not guaranteed quantities.

A. Transportation: The Contractor shall provide adequate and safe methods for trucking and transporting equipment and materials onto and off the projects sites. All Local, State, and Federal regulations and laws shall be complied with by the Contractor during trucking and transporting operations on Local, County, State, and Forest roads.

B. Clean substrate of any projections and substances detrimental to shingling work.

C. Proceed with installation of underlayment and shingles only when weather conditions are in compliance with manufacturer's recommendations and when substrate is completely dry.

1.2 DEMOLITION OF EXISTING ROOF

Temporary Protection: Maintain integrity of attic enclosure keeping all openings, holes, joints, and gaps sealed to prevent entry from bats or other wildlife. Erect temporary protection, such as walks, fences, railings, canopies, and covered passageways, where required by authorities having jurisdiction, and as indicated.

1. Provide protection to ensure safe passage of people around building demolition area to and from occupied portions of adjacent buildings and structures, in accordance with Federal OSHA requirements.

   A. Warning Signs and Labels:

      Provide warning signs at all approaches to the work areas. Locate signs at such a distance that personnel may read the sign and take the necessary protective steps required before entering the area. Warning signs may be in the form of continuous plastic tape. The warning signs shall have black characters on a yellow background.

WARNING
DO NOT ENTER

Alternate wording for the warning signs will be approved by the Contracting Officer.

1.3 DIMENSIONS FIELD VERIFICATION

A. Field verify all dimensions affecting the work of this contract prior to ordering materials. Dimensions provided in the contract are estimates.

C. Due to the age of the structures, the lumber could be dimensional or nominal. Verify prior to ordering material.

1.4 GUARANTEE

A. Furnish a guarantee-warranty in writing against a defective roofing materials for at least 10 years from date of acceptance. Furnish a guarantee-warranty to include repairing if necessary to maintain roof in a watertight condition for 1 year from date of acceptance.

1.5 **Measurement and Payment** - Measurement and payment for contract work will be made only for and under those pay items included in the SCHEDULE OF ITEMS. All other work and materials will be considered as incidental to the pay items shown.

A. When more than one class, size, type, thickness, etc. is specified in the Schedule of Items for any pay item, suffixes will be added to the item number to differentiate between the pay items.
B. Payment shall be by units defined and determined according to U.S. Standard measure and by the following:

   1. Lump Sum (LSQ): One complete unit.
   2. Square (SQ): Measured in 100 SF area of the flat surface of the roof.

C. Method of Measurement: One of the following methods of measurement for determining final payment is designated on the Schedule of Items for each pay item:

1. ACTUAL QUANTITIES (AQ) - These quantities are determined from actual measurements of completed work.

2. LUMP SUM QUANTITIES (LSQ) - These quantities denote one complete unit of work as required by or described in the contract, including necessary materials, equipment, and labor to complete the job. They shall not be measured.

1.6 **All Submittals** - It will be the Contractor's responsibility to submit information on all items required with such promptness as to cause no delay in the work. Approvals must be obtained before starting work on parts of the project which involves the submittal. Contractor shall allow 10 calendar days for processing of submittals by the Government.

A. Submit four copies of submittals and shop drawings to the Contracting Officer for approval on the items required herein. Literature which is not marked or otherwise designated to show the exact product and size that the contractor proposes to use will be returned by the Contracting Officer "without action." Circle or check items to be used on submittals. Highlighter pens may print black writing or may not show up at all when copied and should not be used. One copy of all submittals and shop drawings will be returned to the contractor. Information must be adequate to determine compliance with specifications.

B. **Substitution** - Materials in the specifications may be followed by the words "or approved equal." In these cases, wherever the name or brand of a manufacturer's article is specified, it is used as a measure of quality and utility or a standard. If the Contractor prefers to use any other brand or manufacturer of same quality, appearance and utility to that specified, he shall request substitution as provided below, not less than 10 days before the planned installation of the item. The Contracting Officer will approve or disapprove the request for substitution and his decision shall be final. Unless substitutions are requested within the time stated above and as provided below, no deviation from the specifications will be allowed.

Requests for substitutions will only be considered if contractor submits the following:

1. Complete technical data including drawings, complete performance specifications, test data, samples and performance tests of the article proposed for substitution. Submit additional information if required by Contracting Officer. All items in the above information shall be circled, tagged, or marked in some way to indicate all deviations or differences which the proposed item differs from the originally specified item.
2. Similar data as above for item originally specified. All items shall be marked to identify where/how the proposed substitution will differ.

3. A statement by the Contractor that the proposed substitution is in full compliance with the contract documents, applicable codes, and laws; and that the Contractor takes on the responsiblity for any effect upon related work in the project for any substitution and shall pay any additional costs generated by any substitutions.

1.7 **Related Work** - The work shall be in accordance with these specifications. The following is a list of the specifications that are included with this package.

SECTION 000500 – FENN BUNKHOUSE AND COOKHOUSE ROOFING
SECTION 001900 – MOBILIZATION
SECTION 07720_02 ROOF ACCESSORIES
SECTION 024119 – SELECTIVE DEMOLITION
SECTION 074113-1 – ROOF REPLACEMENT
SECTION 074113-3 – ROUGH CARPENTRY
END OF SECTION
May 2024

SECTION 001900 - MOBILIZATION

PART I - GENERAL

1.1 **Mobilization** - This item is intended to compensate the Contractor for operations including, but not limited to, those necessary for the movement of personnel, equipment, supplies, and incidentals to the project site; for payment of premiums for bonds and insurance for the project; and for any other work and operations which must be performed or costs that must be incurred incident to the initiation of meaningful work at the site and for which payment is not otherwise provided for under the contract.

1.2 **Method of Measurement** - The measurement shall be lump sum for mobilization.

1.3 **Basis of Payment** - The lump sum price shall include full compensation for mobilization. Progress payments for mobilization will be made as follows:

A. Bond premiums will be reimbursed after receipt of the evidence of payment.

B. 50% of the lump sum, not to exceed 5% of the original contract amount, will be paid following completion of 5% of the original contract amount not including mobilization and bond premiums.

C. Payment of the remaining portion of the lump sum, up to 10% of the original contract amount, will be paid following completion of 10% of the original contract amount not including mobilization and bond premiums.

D. Any portion of the lump sum in excess of 10% of the original contract amount will be paid after final acceptance.

Progress payments for mobilization and preparatory work shall be subject to retainage.

PART II - PRODUCTS (Not Applicable)

PART III - EXECUTION (Not Applicable)

END OF SECTION

February 2017

SECTION 07720
ROOF ACCESSORIES

PART 1 - GENERAL

1.1 SUMMARY

A. This Section includes the following:
   A. Materials
   B. Installation

1.2 SUBMITTALS

A. Product Data: For each product indicated.

B. Coordination Drawings: Roof are drawn approximately to scale and coordinating penetrations and roof-mounted items are indicated. Contractor shall field verify dimensions and Penetrations sizes and locations.

C. Samples: For each exposed finish.

1.3 QUALITY ASSURANCE

A. Standards: Comply with the following:

   A. SMACNA's "Architectural Sheet Metal Manual" details for fabrication of units, including flanges and cap flashing to coordinate with type of roofing indicated.
   B. NRCA's "Roofing and Waterproofing Manual" details for installing units.

PART 2 - PRODUCTS

2.1 MATERIALS

A. Aluminum:

   A. Sheet: ASTM B 209 (ASTM B 209M) for alclad alloy 3005H25 or alloy and temper required to suit forming operations, with mill finish, unless otherwise indicated.
   B. Extrusions: ASTM B 221 (ASTM B 221M) alloy 6063-T52 or alloy and temper required to suit structural and finish requirements, with mill finish, unless otherwise indicated.

B. Galvanized Steel Sheet: ASTM A 653/A 653M with G90 (Z275); commercial steel, unless otherwise indicated.

C. Aluminum-Zinc Alloy-Coated Steel Sheet: ASTM A 792/A 792M with Class AZ-50 (AZ-150) coating, structural quality, Grade 40 (Grade 275), or as required for strength.

D. Wood Nailers: Softwood lumber, pressure treated with waterborne preservatives for aboveground use, complying with AWPA C2; not less than 1-1/2 inches (38 mm) thick.

E. Fasteners: Same metal as metals being fastened, or nonmagnetic stainless steel or other noncorrosive metal as recommended by manufacturer. Match finish of exposed fasteners with finish of material being fastened.

F. Gaskets: Manufacturer's standard tubular or fingered design of neoprene, EPDM, or PVC; or flat design of foam rubber, sponge neoprene, or cork.

G. Bituminous Coating: SSPC-Paint 12, solvent-type bituminous mastic, nominally free of sulfur and containing no asbestos fibers, compounded for 15-mil (0.4-mm) dry film thickness per coating.

H. Mastic Sealant: Polyisobutylene; nonhardening, nonskinning, nondrying, nonmigrating sealant.

I. Elastomeric Sealant: Recommended by unit manufacturer that is compatible with joint surfaces; ASTM C 920, Type S, Grade NS, Class 25.

J. Roofing Cement: ASTM D 4586, nonasbestos, fibrated asphalt cement designed for trowel application or other adhesive compatible with roofing system.

K. Aluminum Ridge Caps: Fabricate with baffles to prevent snow and rain entering and with weep holes to allow water to drain to roof. Provide required splice plates and end caps.
   A. Clear Anodic Finish: Class I coating complying with AAMA 607.1.
   B. Color Anodic Finish: Class I coating complying with AAMA 606.1 or AAMA 608.1.
      a. Color: As selected by CO from full range of industry colors and densities.
   C. Baked-Enamel Finish: Manufacturer's standard two-coat thermocured system.
      a. Color and Gloss: As selected by CO from manufacturer's full range.
   D. High-Performance Organic Coating Finish: Fluoropolymer three-coat system with topcoats containing not less than 70 percent polyvinylidene fluoride resin by weight; complying with ASCA 96 or AAMA 620.
      a. Color and Gloss: As selected by CO from manufacturer's full range.

L. Zinc Roof Strip: minimum 3.5 inch wide roof strips to prevent moss and mildew growth. Attach with rubber gasket roofing nails.

2.2 METAL FLASHING AND TRIM

A. Fabricate sheet metal flashing and trim to comply with recommendations in SMAC-NA's "Architectural Sheet Metal Manual" that apply to design, dimensions, metal, and other characteristics of item.

A. Apron Flashings: Fabricate with lower flange a minimum of 5 inches (125 mm) over and 4 inches (100 mm) beyond each side of downslope shingles and 6 inches (150 mm) up the vertical surface.

B. Step Flashings: Fabricate with a headlap of 2 inches (50 mm) and a minimum extension of 5 inches (125 mm) over the underlying shingle and up the ver-tical surface. Or as recommended by roofing manufacturer instructions.

C. Drip Edges: Fabricate in lengths not exceeding 10 feet (3m) with 2-inch (50 mm) roof deck flange and 1-1/2 inch (38 mm) fascia flange with 1/2 inch (9.6 mm) drip at lower edge.

B. Vent Pipe Flashing: ASTM B 749, Type L51121, at least 1/16 inch (1.6 mm) thick. Provide lead sleeve size to slip over and turn down into pipe, soldered to skirt at slope of roof and extending at least 4 inches (100 m) from pipe onto roof.

C. Available Manufacturing Requirements: American made product. Subject to compli-ance with roofing manufacturing requirements.

B. General: Corrosion-resistant flashing must be used to help prevent leaks where a roof meets a wall, another roof, a chimney or other objects that penetrate a roof.

1. Metal Flashing: minimum dimensions need to be 4”(102mm) by 4”(102mm) by 7 5/8’(193mm).

2.3 RELIEF VENTS

A. Available Manufacturers: Subject to compliance with requirements, manufacturers of-fering products that may be incorporated into the Work include, but are not limited to, the following:

A. Aura Ventilation, Inc.
B. Bristolite Skylights.
C. Commodity Products Company, Inc.
D. Dowco Products Group.
E. Dur-Red Products, Inc.
F. Goeller Enterprises.
G. Metallic Products Corporation.
H. Solar Group (The).
I. ThyCurb, Inc.
J. Trimco, Inc.
K. Vent Products Co., Inc.

L. Western Canwell.

B. Low-Profile Gravity Ventilators: Of sizes, style, and profile indicated.

A. Material: Galvanized steel sheet.

a. Prime-Painted Finish: Manufacturer's standard exterior galvanized steel primer specified in Division 9 Section "Painting."
b. Baked-Enamel Finish: Manufacturer's standard two-coat thermocured system.

1) Color and Gloss: As selected by Contracting Officer (CO) from manufacturer's full range.

c. High-Performance Organic Coating Finish: Fluoropolymer **three**-coat system with topcoats containing not less than 70 percent polyvinylidene fluoride resin by weight; complying with **AAMA 621 for galvanized steel** or **ASCA 96 or AAMA 620 for aluminum**.

1) Color and Gloss: As selected by CO from manufacturer's full range.

d. Clear Anodic Finish: Class I coating complying with AAMA 607.1.
e. Color Anodic Finish: Class I coating complying with AAMA 606.1 or AAMA 608.1.

1) Color: As selected by CO from full range of industry colors and densities.

B. Bird Screens: 1/2-inch- (13-mm-) square mesh with 0.062-inch- (1.6-mm-) diameter, stainless-steel wire.

C. Insect Screens: 14-by-18 (1.5-by-1.1-mm) mesh with 0.0123-inch- (0.3-mm-) diameter, anodized aluminum wire in removable, rewirable frames.
D. Manual Dampers: Designed for operation from floor directly below ventilator unit.
E. Roof Curb Construction: Curb-mount units designed for installing 1-1/2-inch- (38-mm-) thick wood curbs.

F. Roof Curb Construction: Self-flashing units with integral self-supporting double-wall aluminum curb, enclosing minimum 1-inch- (25-mm-) thick, glass-fiber board insulation (or equivalent), and with minimum 3-inch (75-mm) roof flanges.

2.4 RIDGE VENTS

A. Available Manufacturers: Subject to compliance with requirements, manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:

A. Air Vent, Inc.
B. Alcoa Building Products.
C. Commodity Products Company, Inc.
D. Cor-A-Vent, Inc.
E. GAF Materials Corporation.
F. Klauer Manufacturing Co.
G. Metallic Products Corporation.
H. Mid-America Building Products Corporation.
I. Niff-Corr, Inc.
J. Obdyke, Benjamin, Inc.
K. Petersen Aluminum Corp.
L. Plyco Corporation.
M. Solar Group (The).
N. ThyCurb, Inc.
O. Trimco, Inc.
P. Trimline Roof Ventilation Systems.
Q. Western Canwell.

B. General: Ventilating ridge cap with ventilating mesh providing a minimum net free area of 18 sq. in./ft. (380 sq. cm/m).

C. Aluminum Ridge Cap: Fabricate from sheet aluminum with baffles to prevent snow and rain entering and with weep holes to allow water to drain to roof. Provide required splice plates and end caps.

A. Clear Anodic Finish: Class I coating complying with AAMA 607.1.
B. Color Anodic Finish: Class I coating complying with AAMA 606.1 or AAMA 608.1.

a. Color: As selected by CO from full range of industry colors and densities.

C. Baked-Enamel Finish: Manufacturer's standard two-coat thermocured system.

a. Color and Gloss: As selected by CO from manufacturer's full range.

D. High-Performance Organic Coating Finish: Fluoropolymer **three**-coat system with topcoats containing not less than 70 percent polyvinylidene fluoride resin by weight; complying with ASCA 96 or AAMA 620.

a. Color and Gloss: As selected by CO from manufacturer's full range.

D. Plastic Ridge Cap: High-density polypropylene or other UV-stabilized plastic designed to be installed under shingles at ridge.

2.5 GUTTERS

A. Gutters shall be 6” Half Round 26 Ga, Galvanized steel with Single Bead, gutter meeting ASTM A653-LFQ. Gutters shall have 90 degree box mitered corners of same gauge and material and half round caps if necessary.

B. Downspouts shall be 4” round 26 Ga galvanized steel with 4” round 26 Ga galvanized elbows, and galvanized gutter outlets.

2.6 DRIP EDGE

A. Available Manufacturing Requirements: American made product. Subject to compliance with roofing manufacturing requirements.

B. General: Metal Drip Edge needs 1⁄2" overhanging the rakes and eaves.

2.7 FLASHING

A. Available Manufacturing Requirements: American made product. Subject to compliance with roofing manufacturing requirements.

B. General: Corrosion-resistant flashing must be used to help prevent leaks where a roof meets a wall, another roof, a chimney or other objects that penetrate a roof.

1. Metal Flashing: minimum dimensions need to be 4”(102mm) by 4”(102mm) by 7 5/8’(193mm).

2.8 MAINTENANCE REPAINTING

A. GENERAL REQUIRMENTS

Available Manufacturing Requirements: American made product. The fascia boards are to be painted before installing the drip edge. Paint should match existing fascia boards.

B. PREPARATORY CLEANING, PAINT REMOVAL, PATCHING, AND REPAIRING

Fascia board needs to be preparatorily cleaned with chemical cleaning or pressure washer. Then loose, hanging, compromised paint needs to be removed with hand tools such as a wire brush. Then begin the patching and repairing to the drip edges exposed sides with substrates with dents and gouges larger than **1/4 inch.**

C. QUALITY ASSURNACE

A. Color Matching: Computer matched.
B. A Material Safety Data Sheet (MSDS) needs to be provided by the paints manufacturer for each color scheme used.

2.9 INSTALLATION

A. General: Coordinate installation of roof accessories with installation of roof deck, roof insulation, flashing, roofing membranes, penetrations, equipment, and other construction to ensure that combined elements are waterproof and weathertight. Anchor roof accessories securely to supporting structural substrates so they are capable of withstanding lateral and thermal stresses, and inward and outward loading pressures.

B. Install roof accessory items according to construction details in NRCA's "Roofing and Waterproofing Manual," unless otherwise indicated,

C. Separation: Separate metal from incompatible metal or corrosive substrates, including wood, by coating concealed surfaces, at locations of contact, with bituminous coating or providing other permanent separation.

D. Flange Seals: Unless otherwise indicated, set flanges of accessory units in a thick bed of roofing cement to form seal.

E. Cap Flashing: Where required as component of accessory, install cap flashing to provide waterproof overlap with roofing or roof flashing (as counterflashing). Seal overlap with thick bead of mastic sealant.

F. Operational Units: Test-operate units with operable components. Clean and lubricate joints and hardware. Adjust for proper operation.

G. Clean exposed surfaces according to manufacturer's written instructions. Touch up damaged metal coatings.

H. Gutters: Every 10’ length of gutter shall be installed with a minimum ¼” drop. Slip joint connectors shall be sealed and riveted. Gutter caps shall be installed using a sealant and crimp/rivet. Brackets shall be installed every 2’ of gutter length to the facia boards. Facia boards must be a minimum of 2” wide for bracket installation. If narrower, use rafter tails.

END OF SECTION 07720

SECTION 024119 - SELECTIVE DEMOLITION

GENERAL

1.1 Items indicated to be removed become Contractor's property. Detach materials designated for removal from existing construction and dispose of the existing roofing materials and items of off National Forest in accordance with all state and federal laws. The Contractor shall be responsible for all costs, royalties, arrangements, procurement, cleanup, and work associated with a waste disposal area.

1.2 Government may occupy portions of building immediately adjacent to selective demolition area. Conduct selective demolition so Owner's operations will not be disrupted. It is not expected that hazardous materials will be encountered in the work. If hazardous materials are encountered, do not disturb; immediately notify CO in writing. Hazardous materials will be removed by Government under a separate contract action.

1.3 Measurement of Payment - SELECTIVE DEMOLITION, will be paid in lump sum quantity and will include all costs of demolition, collection haul, fees, and waste disposal.

PRODUCTS (NOT USED)

PEFORMANCE REQUIREMENTS

Regulatory Requirements: Comply with EPA regulations and with hauling and disposal regulations of authorities having jurisdiction.

Comply with National Historic Places Act NHPA on the demolition and replacement of roofing, by taking care to not damage other parts of the building.

Endangered species act. Protect Brown bats by performing work outside of the occupation window of May through September.

Standards: Comply with ANSI/ASSE A10.6 and NFPA 241.

EXECUTION

WASTE MATERIAL DISPOSAL

1. This item shall consist of loading, handling, hauling, and disposal of excess materials and trash resulting from construction activities. The existing roofing materials and items shall be disposed of off National Forest in accordance with all state and federal laws. The Contractor shall be responsible for all costs, royalties, arrangements, procurement, cleanup, and work associated with a waste disposal area.

2. Measurement of Payment - No separate measurement or payment will be made for disposal of waste material disposal, rather payment will be considered to be incidental to other items of work.

3. All trash and construction debris resulting from installation, demolition, and construction are the responsibility of the Contractor and shall be removed and disposed of legally.

DEMOLITION

A. Maintain services/systems indicated to remain and protect them against damage during selective demolition operations. Before proceeding with demolition, provide temporary services/systems that bypass area of selective demolition and that maintain continuity of services/systems to other parts of the building.
B. Provide temporary barricades and other protection required to prevent injury to people and damage to adjacent buildings and facilities to remain.

C. Protect walls, ceilings, facia, and other existing improvements work that are to remain. Erect and maintain dustproof partitions if interior work is needed. Cover and protect furniture, furnishings, and equipment that have not been removed.

D. Provide and maintain shoring, bracing, and structural supports as required to preserve stability and prevent movement, settlement, or collapse of construction and finishes to remain, and to prevent unexpected or uncontrolled movement or collapse of construction being demolished.

E. Provide temporary weather protection to prevent water leakage and damage to structure and interior areas.

F. Neatly cut openings and holes, keeping work square, plumb and true to dimensions required. Use cutting methods least likely to damage existing construction or adjoining construction.

G. All trash and construction debris resulting from installation, demolition, and construction are the responsibility of the Contractor and shall be removed and disposed of legally. It is the Contractor's responsibility to locate an acceptable disposal area, and to pay all costs involved in the disposal of materials.

H. Clean adjacent structures and improvements of dust, dirt, and debris caused by selective demolition operations. Return adjacent areas to condition existing before selective demolition operations began.

END OF SECTION

May 2024

SECTION 074113-1 ROOF REPLACEMENT

PART 1 - GENERAL

1.1 DESCRIPTION OF WORK

Roof replacement work includes removal of existing wood shingles, ice and water shield, and other materials, selective removal and replacement of damaged or rotten subroof planking, and installation of new roofing Ice and Water Shield, Cedar Breather, and wood shingles. Attic areas must remain sealed throughout entire construction project to prevent entry from bats and other wildlife. The existing roofing materials and items shall be disposed of off forest in accordance with all state and federal laws.

1.2 GENERAL

A. **SCOPE –** Perform all work necessary or required for the removal and replacement of the 16" wood cedar shingle roof in accordance with specification(s) and as indicated on the drawings. Some minor subroofing repairs may be necessary. Final cleanup of the site is included in the work.

B. **Grade Marks –** Each bundle of shingles shall bear a grade stamp showing the grade of the wood and the grading authority.

C. **Materials Handling –** All materials shall be delivered to the site in an undamaged condition, stored in fully covered, well-ventilated areas, and protected from extreme changes in temperature and humidity.

D. **Measurement and Payment –** All work except subroof repair (ROUGH CARPENTRY) will be included in the Lump Sum price for the roof as shown in the Schedule of Items.

Repair of the subroof is not guaranteed and will be paid at the rate shown in the Schedule of Items for the Actual Quantity required and accepted.

1.2 SUBMITTALS

Submit (2) copies to the CO for approval on the items and materials required. It shall be the Contractor's responsibility to submit all items with such promptness as to cause no delay in the work, and obtain CO's approval before starting work that involves the item submitted. Submit information at least 10 working days prior to planned delivery date of product to the worksite.

A. Literature must be marked or otherwise designated to show the exact model, size, etc., the contractor proposed to use. Information must be adequate to determine compliance with specifications.

B. One (1) copy of all submittals will be returned to the contractor.

C. Manufacturer's literature is required for the following items.

1. Shingle Quality
2. Ice and Water shield
3. Cedar Breather or an approved equal
4. Nails
5. Zinc shield
6. Roof Anchors
7. Gutters

1.3 DELIVERY, STORAGE, AND HANDLING

Deliver material to the site, undamaged, with labels intact. Store out of the weather in a well ventilated area. Comply with manufacturer's instructions.

PART 2 – PRODUCTS

2.1 WOOD SHINGLES AND RAKE SHINGLES (Fenn)
Shingles – The shingles shall be 16 inch, No. 1 Red Cedar, Premium Grade, 100 percent V.G. 100 percent clear 5/2 perfect. Treated with cca to prevent moss build up.

2.4 NAILS
Nails for shingle roofing must be Hot Dipped Zinc Coated Nails 6d, size as required to firmly secure the shingles to sheathing. Nails shall not penetrate exposed surfaces. Staples will not be allowed.

2.6 FLASHING
Use 26 gauge galvanized valley and edge flashing.

2.7 SUBROOF SHEATHING/PLANKING AND FASTENERS
Match existing.

2.8 ICE AND WATER SHIELD
Ice and Water Shield – Shall be either Grace "Ultra" Ice and Water Shield as manufactured by W.R. Grace, or "Stormguard HT" as manufactured by Certainteed, or an approved equal may be submitted to the Contracting Officer for review and approval.

2.6 Cedar Breather
Or an approved equal roof ventilation mat compatible with cedar roofing shingles.

2.7 ZINC SHIELD
Zinc shield – Shall be "ZincShield" Zinc strips manufactured by ZincShield or as approved equal may be submitted to the Contracting Officer for review and approval.

2.8 ROOF ANCHORS
Anchors – Shall be French Creek Production, model 1730 or as otherwise approved by the Contracting Officer.

2.9 GALVANIZED STEEL GUTTERS AND DOWNSPOUTS.
Gutters Shall be 6" Half-Round 26 Ga, Galvanized steel with Single Bead, gutter meeting ASTM A653-LFQ. Gutters shall have 90 degree box mitered corners of same gauge and material and half round endcaps as necessary.
Down spouts shall be 4" round 26 Ga galvanized steel with 4" round 26 Ga galvanized elbows, and galvanized gutter outlets.

PART 3 – EXECUTION

3.1 PROTECTION OF STRUCTURE

Take all precautions necessary to protect structure from damage during the roof replacement work.

3.2 SITE PREPARATION

A. All existing roofing materials including roofing felt, shingles, metal flashing, and ridge caps shall be removed and disposed of off forest in accordance with all state and federal laws. The Contractor shall be responsible for any and all charges at a landfill.

B. The existing subroofing shall be cleaned and repaired as necessary, as approved by the CO or COR, with like materials to provide a solid, smooth surface for placement of the new roofing materials.

3.3 ROUGH CARPENTRY

Remove, dispose of, and replace any damaged or rotten subroof sheathing/planks exposed by removal of existing roof. Secure CO approval prior to removing existing roof sheathing. Nail any loose existing sheathing.

3.5 SHINGLE INSTALLATION (Fenn Bunkhouse and Cookhouse)

A. Install ice and water shield in accordance with manufacturer's directions.

B. Install Cedar Breather or approved equal in accordance with manufacturer's directions.

C. Install 3.5" wide Zinc Shield strips entire length of ridgelines, above, and below all points where there is a breach in roofing.

D. Install and use fall protection anchor points in accordance with manufacturer's instruction and OSHA standards 1910.66, 1926.502. Three permanent anchor points

shall be installed per side on the Fenn buildings, with a total of 12 anchor points for the building.

E. The starter course of shingles shall be triple at the eaves with the butts of the starter course projecting 2 inch beyond the first roof board. Install a layer of rake Shingles at the eaves of all dormers and roofs ends. All shingles when laid on the roof, shall be spaced ¼ inch apart. The joints of any one course shall be spread not less than 1½ inch from the joints in the adjacent courses, and the joints alternated so they do not exactly line up over a minimum of three layers. Lay all shingle courses 5.5 inches to the weather. Not more than 2 nails shall be used for each shingle. Each nail shall be placed not more than ¾ inch from the edge of the shingle, and not more than 1 inch above the butt line of the next course, but never so that they are exposed to the weather.
   1.) Hand-nailing the shingles is required. Nails must not be left proud, nor must they be driven below the surface of the shingle. Nails must penetrate at least halfway through the thickness of the sheathing, but must not penetrate completely through the sheathing.
   2.) Only smooth-faced hammers are allowed during this project. No milled-faced or wafflehead hammers are allowed.
   3.) Personnel working on the roof are not allowed to wear spiked or cleated shoes in order to prevent damage to the shingles.
   4.) If the contractor uses a harness and/or safety rigging to conduct roof work, a complete harness and/or safety rigging will be made available by the contractor for Forest Service inspections by the COR and/or the Inspector.

F. Vents stacks extending through the roof shall be made water tight with galvanized flashing with a neophrene ring.

G. Construct ridge caps to match ridge line on the Assessors building (building in front and to the west of the Fenn bunkhouse). Use fasteners, sealants and other miscellaneous items as required to install the ridges to fit the substrate and result in waterproof performance.

3.6 METAL FLASHING INSTALLATION

A. General: Install metal flashings and other sheet metal to comply with requirements in Division 7 Section "Sheet Metal Flashing and Trim."

1.) Install metal flashings according to wood roofing recommendations in NRCA's "The NRCA Roofing and Waterproofing Manual."

PART 4 - MEASUREMENT AND PAYMENT

4.1 GENERAL

The method of measurement will be as designated in the SCHEDULE OF ITEMS.

The accepted quantities will be paid for at the contract unit price for the Pay Items shown in the SCHEDULE OF ITEMS.

Payment will be for all labor, equipment, transportation, supervision, operating supplies, materials, and incidentals necessary to complete the work specified in this section shall be paid for as a Lump Sum.

Repair of the subroof will be paid at the rate shown in the Schedule of Items for the Actual Quantity required and accepted by the CO.

Replacement of gutters will be paid for by the Lump Sum. All work to hang, slope and seal gutters shall be incidental to the item.

END OF SECTION

May 2024

SECTION 074113-3 – ROUGH CARPENTRY

PART 1 - GENERAL

Description of Work - Remove, dispose of, and replace any damaged or rotten subroof sheathing/planks exposed by removal of existing roof. Secure CO approval prior to removing existing roof sheathing. The existing roofing materials and items shall be disposed of off forest in accordance with all state and federal laws.

Grade Marks - All lumber and plywood shall be grade stamped.

Measurement and Payment - No separate payment will be guaranteed for work under this section. Payment for this work must be approved by the CO in writing.

PART 2 - PRODUCTS

Materials - All lumber shall be seasoned, surfaced four sides (S4S), except where replacement in kind deviates from this.

A. **Light Framing** – Cedar, Standard and better grade, as graded by Western Wood Products Association or West Coast Lumber Inspection Bureau, or equal.
B. **Pressure Treated Framing** - FDN grade Douglas fir as designated by American Wood Preservers Association or American Wood Preservers Bureau.
C. **Roof Sheathing and Exterior Wall Sheathing** – APA Rated ¾" sheathing or replace with like kind and dimensions of original material(s), as directed by the CO in writing.
    1. Span Rating: 12/0, 5/16" thick.
    2. Span Rating: 24/0, ½" thick.
    3. Span Rating 24/16, ½" thick.
    4. Span Rating 32/16, ½" or 5/8" thick.
    5. Span Rating 48/24, ¾" or 7/8" thick.
D. **Interior Plywood Wall Sheathing** - CDX fir, interior type, with exterior glue, as graded by the American Plywood Association.
E. **Trim and Moldings - Interior** - C select pine, as graded by the Western Wood Products Association.
F. **Ice & Water Shield** – Grace Ice & Water Shield HT or equal. Install over entire roofing structure.

PART 3 - EXECUTION

General

Nailing - Nailing shall conform with the requirements of the International Building Code, unless otherwise specified. Nails shall be hand driven

General Carpentry

General - All framing shall be accurately cut and constructed plumb, straight, and true to lines and dimensions shown. No cutting and boring of joists and beams shall be done without permission from the Contracting Officer.

Roof Sheathing - The roof sheathing shall be installed with the face grain laid perpendicular to the rafters, end joints staggered, and nailed at each crossing and end. Butted end joints shall occur only over bearing points.

Blocking and Bracing - Blocking needed for bracing, edge support, and backing for miscellaneous accessories, grab bars, toilet paper dispensers, door stops, any needed to stabilize walls and other building elements, and other miscellaneous blocking shall be included as part of this work.

Finish Work

Wood Trim - All wood trim shall be run from solid stock with housed joints. Trim shall be set plumb and securely nailed with all joints tightly fit with no gaps. No gaps in joints which would require putty or other fill will be allowed.

All trim shall be in single lengths. Running finish, such as base and moldings, shall be in commercial lengths and jointed only where solid fastenings can be made. Where end joints are required, miter joints shall be made.

Exterior siding, trim, soffit, and fascia - General: Comply with siding manufacturer's written installation instructions applicable to products and applications indicated unless more stringent requirements apply.
Seal butt joints at inside and outside corners and at trim locations.

END OF SECTION
February 2017